

TABLE I
Relative Data Rates and Range for IEEE 802.11 Standard

Protocol	Release date	Frequency	Typical data rate	Maximum data rate	Indoor range (approx.)
Legacy	1997	2.4 GHz	1 Mbps	2 Mbps	—
802.11a	1999	5 GHz	25 Mbps	54 Mbps	30 metres
802.11b	1999	2.4 GHz	6.5 Mbps	11 Mbps	30 metres
802.11g	2003	2.4 GHz	25 Mbps	54 Mbps	30 metres
802.11n	2006 (draft)	2.4 or 5 GHz	200 Mbps	540 Mbps	50 metres

now called the legacy version. Since then, four more revisions of 802.11 have been released with products developed. Table I shows the basic details of all the protocols. The 'a' and 'g' revisions have the greatest installed base.

Ubiquitous WiFi

The terms 802.11 and WiFi are used interchangeably. 802.11 is the standard and WiFi is the term used to describe wireless LAN (WLAN) products based on the 802.11 standard.

WiFi was originally developed for mobile computing devices such as notebook computers and PDAs, but has found its way into voice-over-IP phones, gaming consoles (Xbox), television sets and digital cameras. It requires the use of access points, which are usually connected via a wired Ethernet connection. Access points broadcast beacons every 100 ms at a data rate of 1 Mbps. This means that the wireless device must be able to operate at a minimum data rate of 1 Mbps. The maximum wireless connections to the access point are determined by the hardware.

Some of the advantages of WiFi are:

1. Significant cost savings in installing LANs in existing buildings
2. Silicon pricing continues to decrease as the WiFi market expands
3. Large number of WiFi products available from access points to client network interfaces
4. WiFi is a global standard and works the same in different countries around the world
5. WiFi is available in your local coffee shop, house, university, business

and many public hotspots

Disadvantages are:

1. Many devices (such as cordless phones and microwave ovens) use the 2.4GHz frequency and other standards like IEEE 802.15.4
2. Channel assignments and limitations are not consistent worldwide. The US has eleven channels, Europe has 13 channels and Japan has 14. Some countries prohibit the use of some channels
3. Most existing devices have fairly high power consumption, making it difficult to integrate into low-power, battery-operated devices. This is changing with devices from Atheros,

which are designed for embedded systems

WiFi in hospitals

Fig. 2 shows the hypothetical setup for a hospital WiFi network. The main use of WiFi-enabled devices in a medical environment such as hospitals would be for patient monitoring and asset tracking. Most hospitals already have a WiFi network deployed, and equipment such as ECG monitors, pulse oximeters, blood pressure monitors and infusion pumps can now implement an IEEE 802.11 transceiver cost-effectively. The information collected can be transmitted to a central server, from which physicians can access the data, and alarms can be generated when irregular conditions are detected. WiFi-based monitors allow patients to be more mobile and make it easier to transfer patients within the various departments in the hospital.

Another use of WiFi-enabled devices is for hospital asset tracking. It is not unusual for hospitals to order more equipment than what is actu-



Fig. 2: Hypothetical setup for a hospital WiFi network. The main Ethernet connection is attached to a wireless router or access point. Devices such as notebook computers, blood pressure monitors, ECG/EKG monitors and infusion pumps can connect wirelessly to the router or access point